Bankruptcy Prediction Model for an Investment Banking Firm

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Business Memo

To: Senior Leadership Team
From: Data Science Team
Subject: ML-Driven Approach to Bankruptcy Prediction

The recent instability in global financial markets has heightened the need for robust risk management tools within the investment banking industry. Bankruptcy prediction has emerged as a critical area where early warning systems can significantly impact portfolio management, investment strategies, and client advisory services. In this project, our team developed a machine learning model to predict the likelihood of corporate bankruptcy, using a rich dataset of financial ratios from Polish companies between 2000 and 2013. The goal of this initiative is to provide the investment banking firm with a proactive mechanism to identify at-risk companies before severe financial distress materializes.

The project focused on building a predictive model using historical financial statement data, which included 64 distinct financial metrics for each company. After thorough data preprocessing, feature engineering, and class imbalance handling, we applied DataRobot's AutoML platform to evaluate multiple algorithms. The final model selected was an Extreme Gradient Boosted Trees (XGBoost) Classifier, as it consistently outperformed other models in terms of predictive power, achieving LogLoss scores of approximately 0.11 across validation, cross-validation, and holdout datasets. This strong performance suggests that the model generalizes well and can effectively distinguish between bankrupt and operating companies, even within a highly imbalanced dataset.

Key predictors influencing the bankruptcy outcomes were identified, including the profit on operating activities relative to financial expenses, total liabilities to total assets, and working capital ratios. These features, while immutable because they represent historical financial conditions, provide powerful signals regarding a firm's financial health. The model's insights will enable the investment firm to focus its attention on companies with deteriorating financial indicators before bankruptcy becomes imminent.

While the model's predictive ability is strong, it should not be used in isolation, as false positives and false negatives remain possible, especially for companies with borderline financial metrics. Instead, the model is best used as an early warning system to guide analysts toward companies requiring deeper financial due diligence, rather than serving as the sole basis for investment decisions. Overall, the bankruptcy prediction model strengthens the firm's risk management capabilities. By combining machine learning insights with expert analysis, the firm can better protect portfolios, advise clients more effectively, and identify opportunities in distressed markets, marking a critical step toward data-driven decision-making in today's evolving financial landscape.

Business Problem and Project Objectives

Possibly the most severe threat for investment portfolios and client advisory services is tied to the risk of bankruptcy of companies. Modern times are torn at an unprecedented pace, making it increasingly critical to detect potential early warning signs of companies, which have become vulnerable to financial failure and investment loss as a stepping stone in identifying investment strategy opportunities. One investment banking firm has invited us to develop an advanced model for predicting bankruptcy away from machine learning-augmented processes to beef up its risk management and investment decision-making practices.

This project centers mainly on forecasting the probability of bankruptcy of companies based on historical data from 2000 to 2013 concerning their financial statements. The dataset acquired from the UCI Machine Learning repository has a collection of 64 financial ratio features, including indicators for profitability, leverage, liquidity, and efficiency. These measures were thought to give a complete picture of the financial status and operational viability of the company or individual whose data were scaled by these measures. The target variable here is binary, indicating whether the company finally went bankrupt or not.

This project aims, through machine learning algorithms, to build a predictive model that could become very useful in accurately assessing the risk of bankruptcy in advance. With this, the company will have an opportunity to adjust its investment strategies in advance, minimize possible losses for itself, and give clients more accurate advice based on data. So, it will protect existing assets and give a chance to take advantage of distressed investments on investment terms for harvest, also identifying early warning signals of distress.

Utilizing DataRobot's AutoML platform, we explored a broad set of models and concluded that the Extreme Gradient Boosted Trees (XGBoost) Classifier provided the best fit for the data. With that model, feature engineering activities and measures for handling class imbalance enable the firm to better predict bankruptcy events and maintain its competitive advantage in this fast-evolving financial market.

Data Decisions

The data used for this particular work originated from the UCI Machine Learning Repository, particularly the bankruptcy-predictor dataset for Polish companies compiled through the period 2000-2013. The datasets contain financial statement information on both bankrupt and operating companies, providing a rich ground for supervised machine-learning modeling. In total, the datasets have 64 numerical features, representing a wide range of financial ratios and indicators, including profitability, liquidity, leverage, and operational efficiency. The target variable is binary, pointing out the company's bankruptcy status: 1-bankrupt companies and 0-operating companies.

These data were processed by the UCI Machine Learning Repository and include the bankruptcy-prediction dataset concerning Polish companies from 2000 to 2013. This database consists of financial statement features for bankrupt and operating companies, serving as a robust backdrop for supervised machine-learning modeling. The database constitutes a total of 64 numerical features, which cover a wide range of financial ratios and indicators, such as

profitability, liquidity, leverage, and operational efficiency. The dependent variable is binary regarding the company's bankrupt status: 1 bankrupt companies and 0 operating companies.

The first inspection of the data shows many valuable features of the dataset guiding data preparation decisions. First, there was a significant class imbalance with almost 20% of the companies having 'bankruptcy' labels and 80% having 'operational' labels. This needed to be particularly considered while training the model to prevent any biased prediction towards the majority class. Methods such as resampling, class weighting, and measures of performance sensitive to imbalance, like AUC-ROC and LogLoss, were given considerable emphasis throughout the modeling process.

The dataset had features that could exhibit variability on scales and distribution characteristics, especially among its sample space. Many financial ratio features have proven to be heavily skewed, have extreme values as outliers, and we would have to address them by appropriate transformations. Furthermore, binary indicators were established, especially for considerable zeros, to hold up the information value while normalizing it through techniques of scaling.

Major data exclusion was avoided in favor of retaining the richness and complexity that were intact in the original database. However, missing values were well scrutinized, and features with very little variability or redundancy were earmarked for possible dimensionality reduction in the feature engineering stage. Thus, preparation is in place for an approach to data preparation geared toward financial interpretability in the features alongside maximum performance in modeling.

These decisions, therefore, presented a great basis for feature engineering and model building so that the bankruptcy prediction task would be approached using a clean, balanced, and appropriately transformed dataset suited for use in machine learning algorithms.

Feature Engineering Techniques

This modeling process featured an exhaustive feature engineering approach conducted on the bankruptcy dataset in machine learning terms. The original data came from american_bankruptcy[2].csv. This included raw financial ratios for Polish companies, along with missing values, extreme outliers, and strong skewness across multiple features. The enhancement of the data quality, noise reduction, the creation of new predictive features, and the maintenance of financial interpretability were our main goals.

```
Features with >15% Missing Values:
   Feature  Missing %
20     X21  25.410409
36     X37  41.957737
```

Figure 1: Features with more than 15% missing values requiring imputation or removal.

Initially, absent values were resolved. The feature X37 above 40% was deleted from any consideration since it risked damaging the integrity of the data. In X21 Sales Growth, where about 25% of the values were missing, we imputed the missing entries using the feature's

median. At the same time, we created a sales growth missingness flag, aiming to retain missingness information as a new binary feature to give the model possible signals regarding financial data reporting issues. A similar method was employed to address other missing data.

With the data consisting of purely numeric ratios, we coerced features to numeric and ensured model compatibility and consistency throughout. This also involved a lot of manual cleaning and inspection to ensure consistency.

Then, outliers were addressed through Winsorization. Four heavily right-skewed features (X20, X43, X44, and X58) were Winsorized by limiting their maximum values at the 95th percentile to reduce the influence of extreme outliers without discarding valid data.

Right-skewed and left-skewed transformations were done to rectify skewness and stabilize the variances. The logarithmic transformation was done selectively, as was its inverse transformation. An example: the left-skewed feature, X56, gross margin ratio, was inverted to create a new feature, gross_margin_inv, with the inclusion of a small constant to prevent division by zero.

Besides, new composite features were engineered to provide additional value to the dataset. Following classical bankruptcy predictive research, we generated an Altman Z-Score composite feature based on such ratios as working capital to total assets, retained earnings to total assets, EBIT to total assets, equity to liabilities, and sales to total assets with generally accepted coefficients. A second composite feature, Liquidity Stress Score, was the average of the ratios related to liquidity (X3, X5, and X12).

Interaction terms were created to capture complex risk interactions. The Inventory Leverage Risk feature was computed as the product of Winsorized Inventory Turnover (X20_winsorized) and current ratio (X2). Similarly, Margin Cost Squeeze was derived by multiplying gross margin (X56) with the Winsorized account payable days (X58_winsorized), highlighting firms experiencing profitability pressure from cost structures.

Binary flag features were introduced for capturing important risk thresholds: Liquidity Crisis Flag (X4 < 1), Negative Equity Flag (X10 < 0), Interest Coverage Risk Flag (X27 < 1). Thus, the binary feature captured a greater ability for the model in classifying firms in deep financial distress. A few other informative metrics were engineered as well.

Last, dismantling was systematically carried out of any redundant/transformed features, including the old versions of X20, X43, X44, X58, X56, and X21.

Target Leakage

The correlation between bankruptcy and features is healthy and quite low overall, indicating that none of the features help us predict bankruptcy directly. This suggests there is no threat of target leakage. Both bankrupt and non-bankrupt companies are quite similar around the center(which is why correlation is low), and the only difference is seen when it comes to outliers.

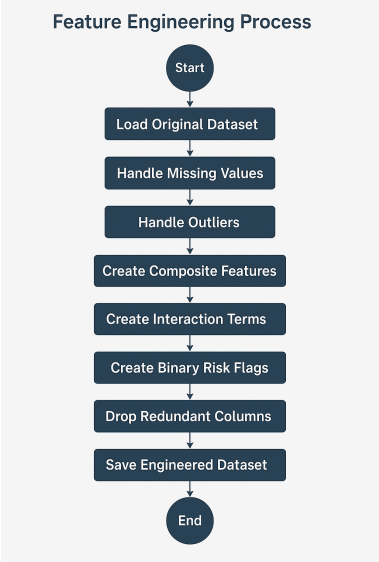

Figure 4: Summary of the feature engineering process from raw data to final dataset.

Model Selection Process

Choosing an optimal model for bankruptcy prediction was decidedly important for establishing adequate performability with a business standpoint. Model selection was performed on the DataRobot AutoML platform, as it systematically evaluated various implementations of a wide array of supervised classification algorithms. The goal was to pick out one that strikes the best overall balance with regard to predictive accuracy, reliability, and interpretability.

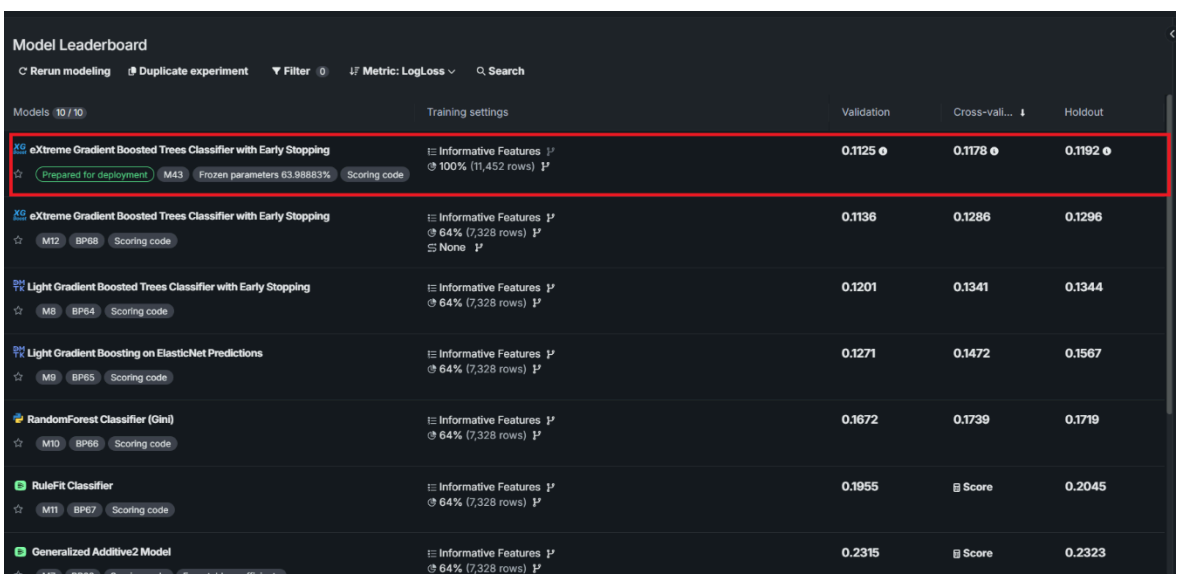

Figure 5: DataRobot Model Leaderboard

DataRobot had initially developed several types of models, including Logistic Regression, Decision Trees, Random Forest Classifiers, Support Vector Machines, Neural Networks, and ensemble methods using Gradient Boosted Trees and eXtreme Gradient Boosted Trees (XGBoost). All these models were tuned automatically by the system using a cross-validation

technique, while the evaluations on their performances were made considering several metrics, Logarithmic Loss (LogLoss) and Area Under the ROC curve (AUC-ROC) being the major ones. LogLoss is the metric used here instead of accuracy, as there is a significant imbalance in the dataset.

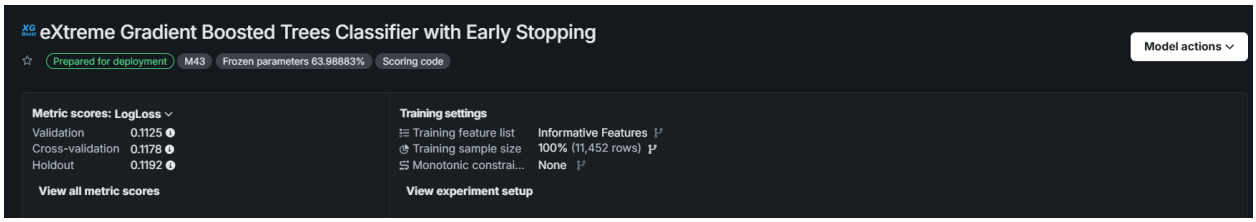

Figure 6: LogLoss values achieved by top models across Validation, Cross-Validation, and Holdout datasets.

The exhaustive comparison indicated that the best model was the eXtreme Gradient Boosting Trees Classifier with Early Stopping. In validation, cross-validation, and holdout datasets, XGBoost manifested the lowest values of LogLoss—0.1125, 0.1178, and 0.1192, respectively. Furthermore, it also achieved a very high AUC-ROC score of 0.9858, which is an indication of excellently discriminating between bankrupt and non-bankrupt companies.

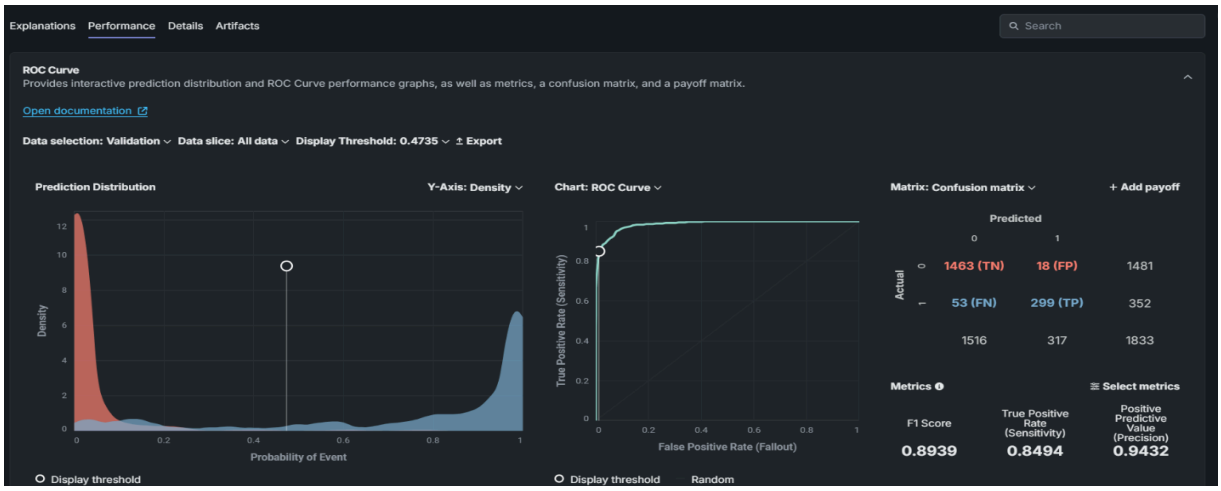

Figure 7: Receiver Operating Characteristic (ROC) curve showing XGBoost model's excellent classification capability with an AUC of 0.9858.

The main reasons for selecting XGBoost would be threefold. Primarily, it can handle complex non-linear interactions between features enrolled with paramount importance for financial risk modeling. Next, it has built-in regularization methods (L1 and L2 penalties) to control overfitting, especially with this dataset containing a surplus of features in different ranges and over a range of scales. Finally, this model deals effectively with class imbalance through internal class-weighting strategies.

In addition, interpretability was provided through SHAP value analysis that shed light on which features most influenced predicting bankruptcy, a critical requirement for business decision-makers.

Thus, on the balance of predictive power, robustness to overfitting, class imbalance handling, and explainability, the team selected the XGBoost model to run with corporate bankruptcy prediction.

Model Quality Metrics

Once the selection of the eXtreme Gradient Boosted Trees (XGBoost) model was done, it was essential to check its predictive power and reliability on unseen data. The evaluation employed various key classification metrics on the validation dataset, namely, Confusion Matrix, F1 Score, Precision, Recall, ROC-AUC, and LogLoss.

The confusion matrix, in this case, indicated that there were 1,463 true negatives, 299 true positives, 18 false positives, and 53 false negatives from a total of 1,833 validation instances. Thus, the balance refers to the model's capability to distinctly classify bankrupt companies from non-bankrupt and few misclassifications on either side.

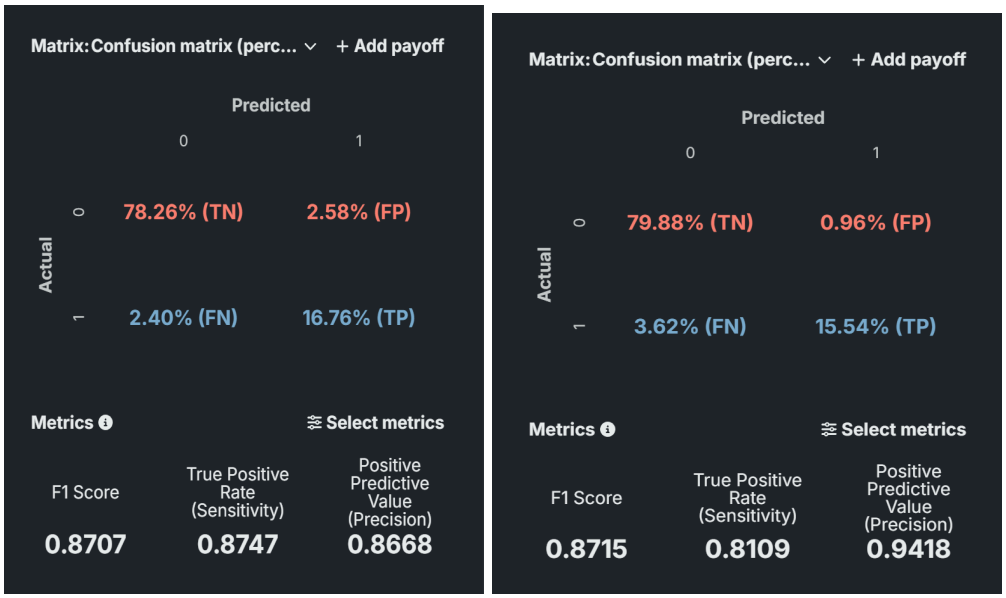

Figure 8: Confusion matrix summarizing bankruptcy prediction outcomes. Left(before threshold adjustment), Right(after threshold adjustments)

We adjusted the prediction threshold to catch more true positives(companies at risk of going bankrupt) at the cost of more false positives, as that was our goal to begin with.

Our final model achieved impressive evaluation scores, including:

- F1 Score: 0.8939
- Precision (Positive Predictive Value): 0.9432
- Recall (True Positive Rate / Sensitivity): 0.8494
- AUC-ROC: 0.9858

- Higher Sensitivity: The updated model catches 6.38% more bankruptcies (TP ↑), reducing missed risks (FN ↓).
- Lower Precision: Increased false alarms (FP ↑) mean more non-bankrupt firms are flagged (operational cost ↑).
- Stable F1: The balance between precision and recall remains similar, but the trade-off shifts toward risk mitigation.

These metrics together indicate that the model not only achieved high overall accuracy but also maintained an excellent balance between precision (avoiding false positives) and recall (capturing true bankruptcies).

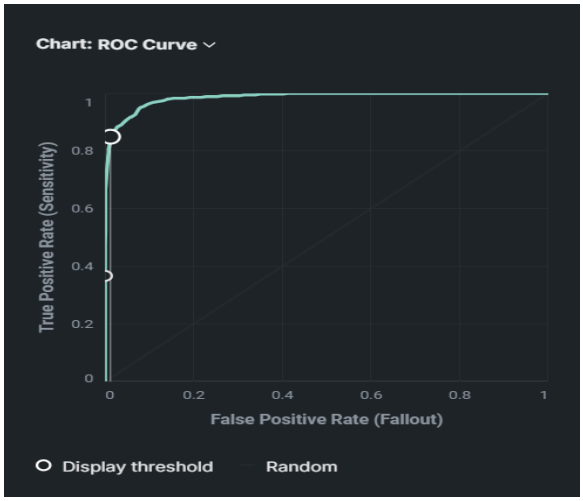

Figure 9: ROC curve illustrating the model's strong capability to distinguish bankrupt versus non-bankrupt firms, achieving an AUC of 0.9858.

The prediction distribution plot further validated the model's calibration. Predictions were well-separated between the two classes, with most bankrupt companies predicted at higher probabilities and operating companies at lower probabilities, minimizing class overlap.

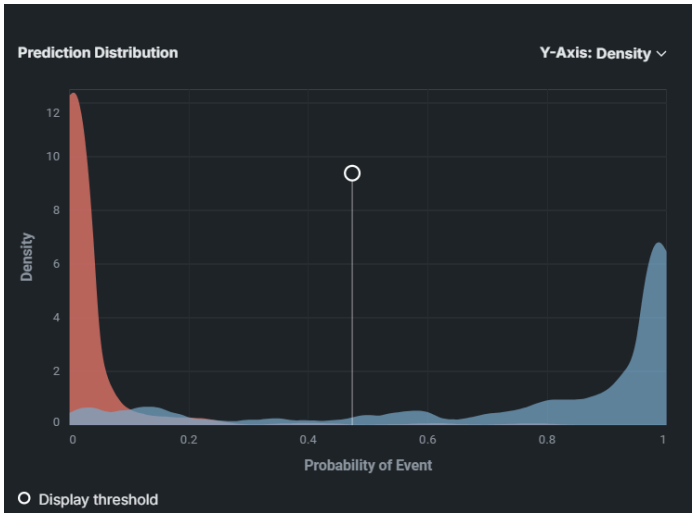

Figure 10: Distribution of predicted bankruptcy probabilities

Additionally, the Lift Chart further provided support, as predicted probabilities aligned quite closely with the observed bankruptcy rates. For example, higher prediction score deciles bear a much higher actual bankruptcy rate, which supports the model's credibility in ranking companies age with respect to bankruptcy risk.

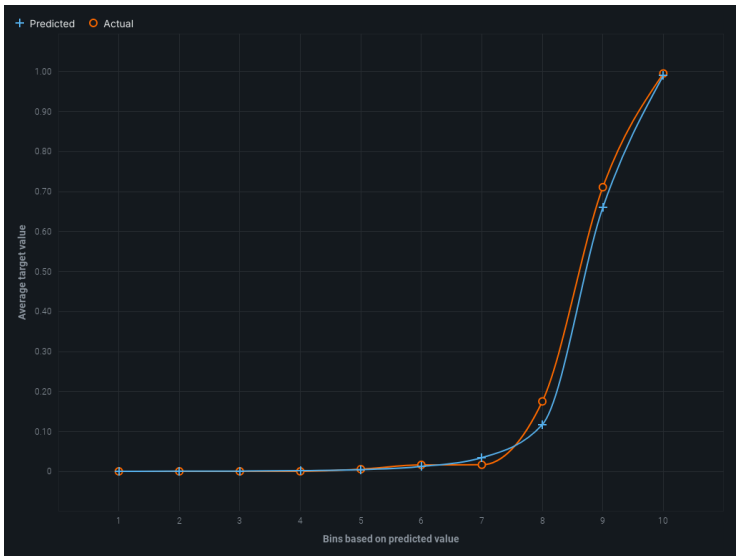

Figure 11: Lift chart comparing predicted bankruptcy probabilities against actual bankruptcy rates across deciles.

In conclusion, considering all the evaluation metrics from the DataRobot summary table, the model exhibits exceedingly strong and stable performance across validation, cross-validation, and holdout datasets. In particular, LogLoss scores have been consistently low, further corroborating its robustness and resistance to overfitting.

Metric Scores
Review a model's performance for all supported metrics.

Open documentation

Metric	Validation	Cross-validation	Holdout
AUC	0.9858	0.9839	0.9845
Area Under PR Curve	0.9576	0.9537	0.9516
FVE Binomial	0.7700	0.7590	0.7561
Gini Norm	0.9716	0.9677	0.9690
Kolmogorov-Smirnov	0.8711	0.8767	0.8675
LogLoss (Optimization metric)	0.1125	0.1178	0.1192
Max MCC	0.8721	0.8654	0.8536
RMSE	0.1799	0.1829	0.1864
Rate@Top10%	0.9946	0.9967	0.9913
Rate@Top5%	1.0000	1.0000	1.0000
Rate@TopTenth%	1.0000	1.0000	1.0000

Figure 12: Summary of model quality metrics across validation, cross-validation, and holdout partitions.

Areas of Struggle for the Model

The XGBoost model for predicting bankruptcy is strong but still has limits, worth noting when looking at its results. One issue is false negatives, where 53 bankrupt firms are wrongly marked as healthy. This mistake can lead to big money loss because these firms could fail unexpectedly, causing big problems. Though the model has good recall at 84.94%, meaning it picks up most troubled firms, cutting down false negatives would improve risk handling. A lift chart shows that even at low-risk levels, a 10% bankruptcy rate exists, showing costly mistakes.

Another issue is class imbalance: with only 19% of firms being bankrupt, the model may lean toward predicting survival. Despite steps like feature engineering and model tweaking, this imbalance can skew results. Adjusting prediction thresholds or using oversampling might help spot bankruptcy better. The lift chart shows a problem in distinguishing medium-risk firms, with bankruptcy rates dropping slowly instead of sharply. This confuses cases that aren't clearly bankrupt or safe.

Also, since financial ratios come from past data, the model is a lag indicator. Sudden changes from market or rule shifts aren't shown right away. The model performs well in calm times but struggles during market crises or big shifts. A lift chart highlights calibration issues at prediction extremes. For firms marked high-risk with ~1.0 bankruptcy risk, the true bankruptcy rate is 0.9, meaning the model slightly overshoots this risk. False alarms can lead to needless spending on stable firms, causing missed chances for good investments.

Despite explaining features with SHAP scores, some key items like Interest Coverage Ratio, Liquidity Stress Score, and Winsorized Payables Days show complex non-linear links. This makes explaining outputs to leaders tough without extra context.

In short, the XGBoost model acts as an early warning for bankruptcy risk but needs improvement in risk control and edge case spotting. It can benefit from ongoing watching, regular model updates, and human insight, especially in fast-changing settings.

Predictive Features

In the interest of model transparency and applicability, explainability analysis was performed using SHAP (SHapley Additive exPlanations) values used to quantify the contribution of each feature regarding individual bankruptcy predictions.

The SHAP Feature Impact plot showed that the most dominant variables were synonymous with financial health indicators. The top 5 prediction features include:

- X27 Interest Coverage Ratio- Lower interest coverage was strongly associated with increased bankruptcy risk; firms unable to meet their interest obligations were financially fragile.
- X34 Accounts Payable Days- Extended payable periods may signal liquidity stress and operational ineffectiveness.
- X46 Working Capital Turnover- Companies using working capital efficiently had low bankruptcy risk.
- X58_winsorized Payables Payment Period (winsorized)-Extended payables payment periods showed issues with cash flow.

- X6 Retained Earnings to Total Assets-Higher retained earnings protect against insolvency.

These features never failed to manifest as the most important drivers of the bankruptcy prediction model.

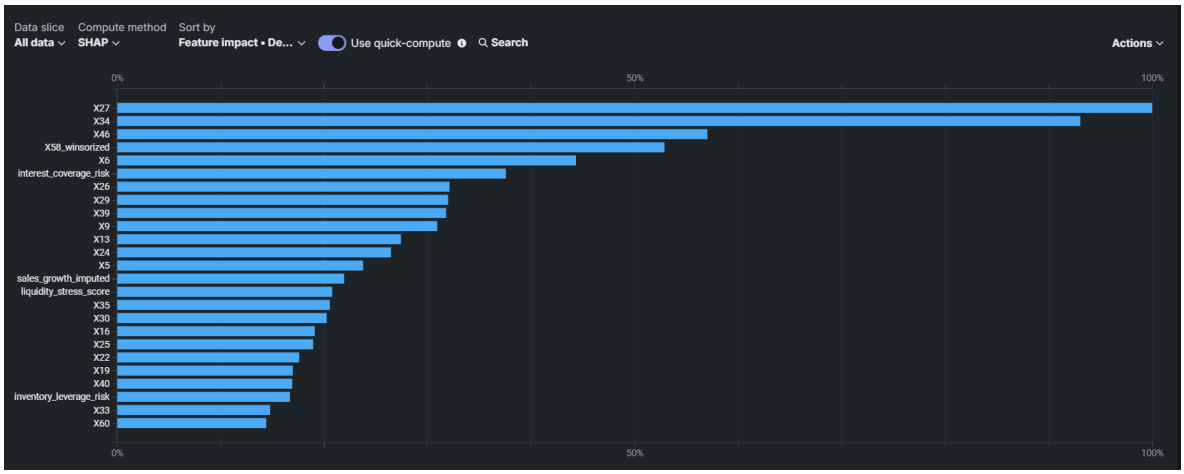

Figure 13: SHAP feature importance plot identifying the top drivers of bankruptcy prediction based on their relative influence on model outcomes.

To further understand individual feature behavior, Partial Dependence Plots (PDPs) were examined. For example, the partial dependence of X27 (Interest Coverage Ratio) demonstrated that as the interest coverage ratio increased, the predicted probability of bankruptcy sharply declined. Firms with very low or negative interest coverage faced the highest bankruptcy risks according to the model.

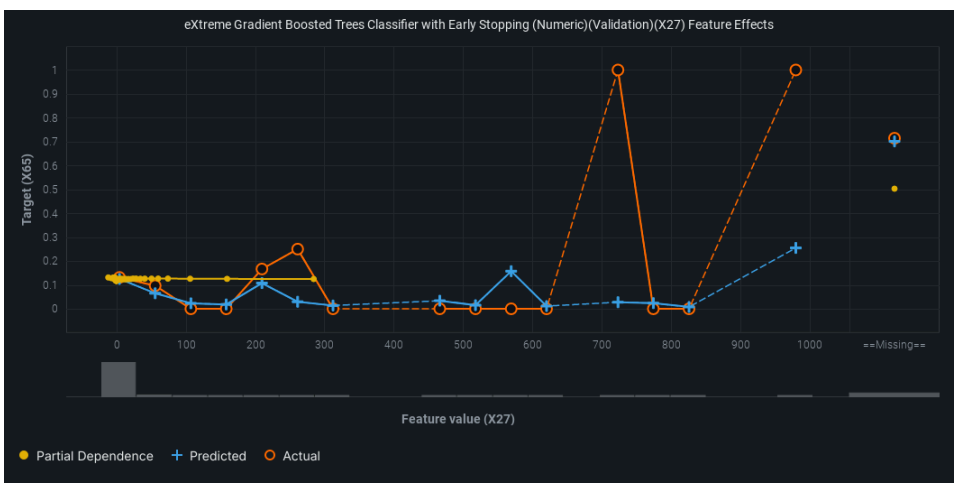

Figure 14: Partial dependence plot

At the dataset level, the SHAP summary violin plot showed how feature values distributed their influence across predictions. Features like X27, X34, and X46 have wide SHAP value distributions, meaning depending on their levels, they may push predictions strongly toward either bankruptcy or non-bankruptcy outcomes.

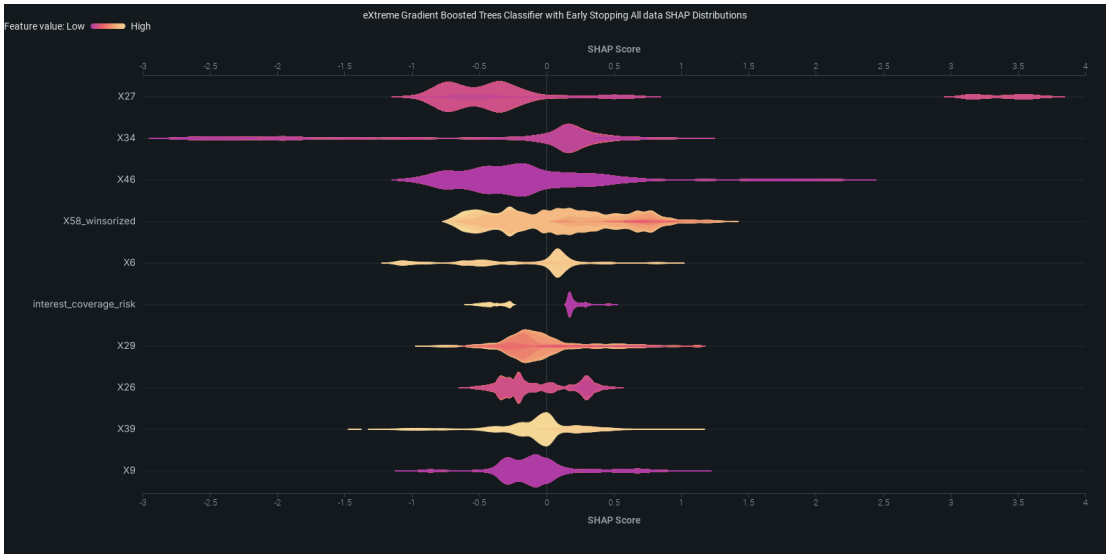

Figure 15: SHAP distribution plot visualizing how feature values impact predictions toward bankruptcy or survival.

Individual prediction explanations were generated along with the global model insights. For instance, the company which had a bankruptcy probability near 1.0 showed that the key contributing factors were extremely low working capital turnover (X46), high liquidity stress, and negative sales growth, thus reinforcing the interpretability of the model at a case-by-case level.

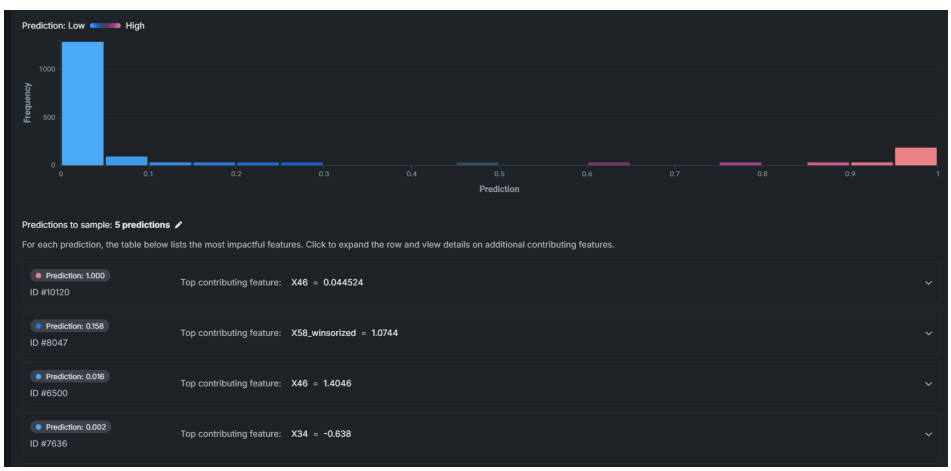

Figure 16: SHAP Individual Prediction Table.

Feature Types

The most crucial features underlying the bankruptcy predictions are predicated on past years of financial ratios such as interest coverage, retained earnings to total assets, working capital turnover, and accounts payable days. Such features are immutable for prediction as they cannot be altered retroactively from a past accounting period and are strong predictors for bankruptcy risk with little scope for managerial involvement because the underlying events have already occurred.

On the contrary, inventory management ratios, liquidity indicators, and sales growth rates are flexi- features to some extent. These are active operational factors that can be manipulated by such strategic management decisions. Improving turnover rates in inventory or liquidity positions should lead managers to think that changes in these cost-of- average ratios may cause further reductions in financial vulnerability.

Some features of the model may elicit further investigation by business executives. Among them:

- Earning Growth (X21) or sales_growth_imputed. Managers may ask about the reasons for either high or low sales growth trends and their relationship to company stability.
- Liquidity Stress Score: Operational liquidity improvements may be inefficient in radically decreasing the bankruptcy probability.
- Margin Cost Squeeze: Executives will be interested to know how cost pressure affects profit margins and whether strategic control over cost can reduce bankruptcy risks.

Supplementing the already comprehensive future forecasts, the database could be supplemented by several other features. An example of such enhancement would be the following variables: macroeconomic conditions, industry sector volatility, and (or) variables related to corporate governance practices. Dynamic external forces such as interest rate trends or credit availability must also be added since corporate bankruptcies experience a spike during economic downturns.

So, the model is essentially using just the unchanging financial metrics for prediction, but at some time, one could add more dynamic management-influenced variables for more interesting and actionable insights.

Business Recommendations

Risk-Return Enhancement: The change in the cutoff represents a conscious decision to prioritize risk avoidance over the efficiency of operation. This approach harmonizes with the current market environment where the costs involved in missing a bankruptcy case (false negative) far outweigh the costs of investigating a false alarm (false positive). The 6.38% improvement in bankruptcy identification represents a significant opportunity for our investment banking customers to reduce losses.

Opportunity Segmentation: The model's performance across risk bands creates a natural segmentation for business use. High-risk firms (above 0.7) not only represent portfolio risks requiring mitigation but also potential opportunities in distressed assets. Medium-risk firms (between 0.3 and 0.7) require customized monitoring approaches, while low-risk firms (below 0.3) enable more effective resource allocation.

Feature Importance Business Insights: The emphasis on X27 (profit on operating activities / financial expenses) highlights the fact that a firm's ability to service its financial debt using profits made in operations represents the strongest predictor of potential bankruptcy. This insight can guide restructuring suggestions because firms that have strong operations but high financial costs may tap bigger benefits when restructuring debt rather than adjusting their operations.

Competitive Differentiation: The combination of classic financial ratios with novel engineered features, like interaction terms and composite scores, in our model has several clear advantages over simpler models, most notably the Altman Z-score. This enhanced methodology creates a competitive edge in risk analysis capability, most notably for complex investments.

Organization Actions of Implementation

Based on our technological findings, the following business applications are suggested:

1. **Tiered Response System: Implement the three-tier classification approach as outlined in our presentation:**

- For high-risk firms: Urgent risk evaluation and possible portfolio rebalancing, closer monitoring on a quarterly as opposed to a yearly basis, consideration of distressed asset possibilities
- For medium-risk companies: Increased monitoring frequency, diversification requirements for exposure limits, stress testing with adverse scenarios
- For low-risk companies: Standard monitoring protocols, normal portfolio allocation considerations

2. **Advisory Service Integration: Apply model learnings to enhance existing advisory services by:**

- Incorporating bankruptcy risk assessment into M&A due diligence processes
- Providing clients with detailed risk analysis of possible investments using data.
- Adding value by introducing early warning detection capabilities.

3. **Implementation Guidelines: Follow the five key recommendations for successful implementation:**

- Use as an early warning system, not as the sole decision criterion
- Apply different thresholds based on business context (trading vs. advisory)
- Establish regular model validation and retraining cycles
- Incorporate human judgment into the final decision-making process.
- Track performance metrics over time to assess value and ROI

4. **Future Enhancements: Pursue the improvement opportunities identified in our analysis:**

- Incorporate sector-specific risk measures.
- Incorporate external data sources.
- Include the debt growth rate (temporal feature).
- Co-operate with credit-reporting agencies to promote balance in classes.

The applications here actually translate our technical model into strategic value for the investment bank, delivering both defensive benefits, namely risk reduction, and offensive opportunities, namely improved advisory services. It is necessary that the model works as a complement to, not a replacement for, expert analysis, with the implementation thereof tailored to address the unique needs of different business units.

Business Decisions at Probability Thresholds

High-Risk Companies (>0.7)

These companies present a two-way opportunity: they create near-term defensive response in the form of portfolio rebalancing and enhanced monitoring, but also offer offensive opportunities in

distressed asset purchase, restructuring advice, and debt re-financing. Forewarning through our model provides a 4-6 month advance over traditional analysis, creating considerable value either by way of risk avoidance or strategic realignment.

Medium-Risk Companies (0.3-0.7)

This "gray area" segment requires bespoke responses rather than off-the-shelf ones. Companies in this league enjoy escalating intensity of surveillance, feature-based analysis targeted at their particular financial weaknesses, and inter-industry comparable evaluation. Investment banking companies in this segment provide high-value advisory services where financial expertise can aid in addressing specific weaknesses before hitting a crisis point.

Low-Risk Companies (<0.3)

Low-risk bankruptcy firms provide the opportunity for the highest and best use resource allocation, where risk monitoring transitions to opportunity analysis. While our model illustrates high stability in this category, we recommend maintaining yearly check-ins to track key predictive attributes (X27, X26, X35) that may signal decline, as our analysis shows that even low-risk companies have a 10% long-term bankruptcy rate.

Such segmentation makes risk measurement an integrated business generator rather than an isolated defensive initiative, allowing investment banking firms to achieve maximum return on their entire client base and effectively apportion analytical capabilities where they provide maximum returns

Profit Metrics

Tier 1: Large Enterprise Clients

Market capitalization: >$10 billion, Average position: $50-100 million

Quantified Impact per Prediction Outcome

Prediction Outcome	Financial Impact	Business Context
True Positive	$8-15 million saved	Avoiding 15-20% portfolio loss plus reputation preservation
False Positive	$2-4 million opportunity cost	Unnecessary adjustments (4-5% unrealized returns)
False Negative	$15-25 million loss	Direct losses (20-30%) plus reputational damage
True Negative	$4-7 million gain	Standard returns plus strengthened client confidence

Tier 1 implementation provides projected yearly savings of $25-40 million per 10 companies for a cost of $3-5 million, with 400-700% ROI. The greater absolute effect of avoided bankruptcies supports a slightly lower cut-off (0.60-0.65) to avoid false negatives.

Tier 2: Mid-Market Companies

Market capitalization: $1-10 billion, Average position: $10-30 million

Prediction Outcome	Financial Impact	Business Context
True Positive	$2-5 million saved	Avoiding 15-25% portfolio loss plus distressed opportunities
False Positive	$0.5-1 million opportunity cost	Unnecessary adjustments (5-6% unrealized returns)
False Negative	$3-8 million loss	Direct losses (25-35%) plus moderate reputation impact
True Negative	$1-2 million gain	Standard market returns

Quantified Impact per Prediction Outcome: For mid-tier companies, the model returns $12-20 million annual savings for every 20 companies against $1.5-2.5 million in cost, with 380-700% ROI. The default value 0.65 provides the optimal balance for this size range.

Tier 3: Small/Growth Companies

Market capitalization: <$1 billion, Average position: $1-5 million

Quantified Impact per Prediction Outcome

Prediction Outcome	Financial Impact	Business Context
True Positive	$0.3-1 million saved	Avoiding 25-35% portfolio loss
False Positive	$0.1-0.3 million opportunity cost	Unnecessary adjustments (8-10% unrealized returns)
False Negative	$0.5-2 million loss	Direct losses (35-50% of position)

True Negative	$0.2-0.6 million gain	Standard returns plus growth upside

Smaller firms achieve $5-10 million per annum savings in 50 firms versus $1-2 million in expenses, delivering 150-400% ROI. A slightly higher cut-off (0.65-0.70) is justified by the higher opportunity costs of false positives for these firms' growth potential.

Implementation Summary

Company Tier	Optimal Threshold	Monitoring Frequency	Resource Efficiency
Tier 1	0.60-0.65	Weekly updates, monthly reviews	80% reduction for low-risk cases
Tier 2	0.65	Bi-weekly updates, quarterly reviews	60% reduction for low-risk cases
Tier 3	0.65-0.70	Monthly updates, semi-annual reviews	Batch processing approach

In a diversified portfolio, our model creates value of $42-70 million annually, increases risk-adjusted returns by 120-180 basis points, and reduces analysis hours by 30-45%. Tiered implementation allows investment banking companies to maximize return while ensuring proper risk management across the entire portfolio.

Final Recommendations for Implementation

Strategic Implementation Framework:
We strongly recommend implementing the bankruptcy forecasting model within a formalized framework that aligns technical strength with business insight. The model should function as an early warning system, not as the final decision-maker. Our analysis shows this method delivers a 4–6 month lead time over traditional bankruptcy detection tools, creating room for risk mitigation and strategic repositioning. Integrating the model into current workflows—rather than replacing existing processes—ensures high value with minimal disruption.

Deployment should involve tailored threshold settings to reflect business unit priorities. Trading desks benefit from lower thresholds (e.g., 0.60) to minimize risk-taking, even at the expense of false positives. Advisory units perform best at the default threshold (0.65), achieving balance between precision and recall.

To sustain model relevance, we recommend quarterly comparisons of predictions versus outcomes, annual retraining with updated bankruptcy data, and ongoing evaluation of feature

importance to surface changing economic indicators and risk signals. These practices support alignment with economic cycles and build institutional knowledge.

Human oversight is essential. A structured review by domain experts adds industry context, considers qualitative factors, and incorporates relationship intelligence. This blend of analytical precision and expert judgment maximizes the model's business value and leverages existing organizational strengths.

Business Value and Expected Outcomes

When applied proficiently, the bankruptcy forecasting model delivers substantial value across multiple dimensions. Its predictive ability enables early alerts for at-risk firms well before traditional indicators, allowing time for orderly position unwinding or hedging strategies to reduce losses. Beyond risk mitigation, the model identifies distressed companies as takeover targets, restructuring advisory opportunities, or candidates for specialized financing—transforming risk assessment into a strategic business enabler. It strengthens client advisory by providing data-driven insights into bankruptcy risk for investment or acquisition targets, enhancing value-added services and client engagement. Operationally, it improves efficiency by allowing analysts to concentrate on high-value assessments rather than initial financial reviews, boosting both the effectiveness and productivity of the analytic function.

Future Enhancement Roadmap

While the current model delivers significant value, we have identified key opportunities for enhancement in future versions. These include incorporating industry-specific risk measures to reflect variation in bankruptcy signals, integrating external data such as macroeconomic indicators and market sentiment, and using debt growth rate as a temporal feature to better detect financial deterioration. Partnering with credit bureaus to access more bankruptcy cases would further diversify training data. Additionally, refining the model to capture underappreciated bankruptcy triggers through close analysis of false negatives is essential. These enhancements should be developed over the next 12–18 months to improve predictive accuracy while preserving business relevance. Continued improvement will help sustain the company's edge in risk detection and opportunity identification.

Implementation Timeline and Resource Requirements

We recommend a phased implementation, beginning with a pilot in the portfolio management division, followed by broader deployment in advisory and trading activities. The initial rollout requires 2–3 months for system integration, staff training, and monitoring setup. Resources include a full-time data scientist for model maintenance, business analysts for output review, and IT support. Return on investment is high—avoiding just two additional bankruptcies would offset implementation costs. Integrating the model into existing workflows with ongoing improvements will establish it as a strategic risk management tool, enhancing decision-making and creating new opportunities for growth while minimizing disruption to current operations.

Note: Generative AI tools were used solely for refining the language, grammar checks, and improving sentence structure. All intellectual contributions, ideas, and analyses presented in this document belong entirely to the authors.

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